

Quasaria perps desks — strategy research (backtest / simulated only)

Prepared for Robert Walker, Oct 04, 2026. Branch *strat/research* (research only; nothing deployed, live site / demo defaults / testnet fleet untouched). All figures are simulated backtests on historical XLM-USD prices; they are not forecasts and imply no future returns.

1. Headline

No configuration we tested is robustly net-profitable out-of-sample after realistic vault costs. We evaluated **4,196 parameter configurations** (11 strategy families × 6 timeframes, 1m → 4h) plus the 10 current demo/fleet desk configs and 36 published-default checks, on up to 2 years of real XLM-USD data, using walk-forward optimisation and a final untouched holdout (2026-08-23 → 2026-10-04, 42 days).

- **Sub-30-minute frames (where the Active demo desks run) are structurally unprofitable:** only 0% (1m), 0.1% (5m) and 0.9% (15m) of configs made money after costs, while ~35-55% were positive before costs — i.e. roughly coin-flip signals, and the ~0.2% round-trip cost (10 bps open fee + 2 × 5 bps slippage; the vault has no close fee) is 1.6× a typical 1-minute bar range and ~3× a 15-second bar range.
- **All six current Active desk configs lose money out-of-sample and in the holdout** (in 22 of 23 OOS windows; the exception is one +\$14 Regal window) (e.g. Vega 5m trend −\$239 over 90 OOS days and −\$168 in the 42-day holdout on a \$500 desk).
- **Positive walk-forward results are almost all 30m-4h trend/momentum** (12 of 65 family×frame combos positive OOS; the sizeable ones are 1h/4h Turtle, TSMOM, Supertrend and trend), but for **all eight top candidates** the profit comes from one or two trending windows (Jul-2025 and May-2026 XLM rallies): removing the single best window turns all of them negative, none survives a multiple-testing correction (deflated-Sharpe probability ≈ 0), and **all eight top walk-forward candidates lost money on the untouched holdout** (−\$1 to −\$77).
- Closest-to-viable, unconfirmed: the textbook **Supertrend 10×3 on 1h** (no optimisation): +\$316 over 480 OOS days (238 trades, PF 1.54, p = 0.005 vs random entries) but **−\$6.9 on the holdout** (21 trades); and the **existing fleet 4h trend default** (+\$165 OOS, −\$36 holdout, 9 trades / 0 wins). These are candidates for forward paper-trading, not for claims of profitability.
- **Security finding (not a strategy):** because the vault fills at the oracle price, if the oracle updates only every ≥2 minutes anyone watching the live market can open when the market is >0.25% away from the stale oracle and close at the next update: simulated +12-16 bps per trade net of all costs with a 64-77% win rate (+39-59 bps at a 0.5% gap). This is an **oracle-latency vulnerability** of the vault and should be fixed, not exploited (§8).

Recommendation: do not present any desk as profitable. If the demo should lose less, move desks to 1h/4h frames (cost drag ~0.1-0.2 ATR instead of 1.6-3 ATR) and retire the skew proxy; see §9 for proposed per-desk settings (research proposal only — not applied).

2. Data

- **Coinbase Exchange public candles, XLM-USD** (cached under *data/*): 1m: 64,801 bars 2026-08-20 → 2026-10-04, 5m: 57,601 bars 2026-03-18 → 2026-10-04, 15m: 38,401 bars 2025-08-30 → 2026-10-04, 30m: 19,200 bars 2025-08-30 → 2026-10-04, 1h: 17,521 bars 2024-10-04 → 2026-10-04, 4h: 4,379 bars 2024-10-05 → 2026-10-04. Missing minutes (no trades) forward-filled.
- **Funding:** Kraken Futures PF_XLMUSD hourly funding history (2025-10-01 → 2026-10-04, 8,840 hours) applied to open positions, capped at the vault's 0.05 %/h. Mean −0.0002 %/h, 95th percentile |0.004 %/h| — economically negligible, so there is no funding carry to harvest and the "skew proxy" desks are not funding strategies. Binance and Bybit public APIs are geo-blocked from the research box; OKX (8-hourly) was reachable but not needed.
- **15s / 30s frames:** no sub-minute history is cached (Coinbase only gives sub-minute data via raw trades, ~hours deep). Echo/Halo (15s/30s mean-rev) are therefore evaluated at 1m with their Active parameters. Costs relative to bar size are 2-3× worse at 15s/30s than at 1m (table in §7), so 1m results are an optimistic bound.

3. Backtester and parity with the fleet

research/engine.py is a numba event-driven single-desk simulator that mirrors *frontend/src/lib/demo/engine.ts* and *bot/src/office/{sizing,risk}.ts*: decisions on closed bars at the close (= oracle price); entry fill = price ± 5 bps; 10 bps open fee on notional; **no close fee** (verified in *contracts/leverage-vault*: only *open_fee_bps*); exits ± 5 bps, take-profit exact; stop first if stop and target are hit in the same bar; gap through a stop fills at the bar open (more conservative than the demo); stop clamped to 0.8-7.5 %; stop-distance sizing with 0.25 % fee buffer, 1-2 % risk (fleet: trend/mean-rev 1 %, funding desks 0.75 %), 1.5× equity notional cap, 25 % margin cap, desk leverage caps (5×/4×/3×), half-liquidation-distance rule, 10-unit minimum margin; desk gates kept: 3 % daily loss, 10 % drawdown halt (+flatten), 5-loss 4 h pause, 6 entries/desk/24 h, max 2 open. Research-only modelling choice: after a 10 % drawdown halt the desk is reset by an "operator" after 7 days (halts are counted in the tables). Not modelled: fleet-wide caps (a single desk at 1 % risk does not bind them), drift halts, oracle-stale entry blocks.

Parity vs the real TypeScript demo engine (same cached 3-day window, 1m stepping, \$500 single desk; *frontend/scripts/parity-backtest.ts* drives the actual *step()*):

desk	ts_trades	py_trades	same_entry_time	ts_net	py_net
vega	18	18	18	-25.48	-29.58
rigel	16	15	12	-28.97	-32.32

desk	ts_trades	py_trades	same_entry_time	ts_net	py_net
lyra	18	18	18	-12.59	-12.59
nova	18	18	18	-15.23	-15.23

Lyra and Nova match exactly (same 18 entries, identical P&L). Vega has identical entries; the P&L gap most likely comes from the fleet-level gross-notional cap that trims the second concurrent position in the TS engine (not modelled for a single desk). Regal differs on 4 entries because the TS indicators are recomputed on a 300-bar rolling window (EMA/ATR seeding) while the Python engine uses the full series. Conclusion: same signals, same costs, same sizing; differences are second-order and do not change any conclusion. (The original `active-backtest.ts` runs all six desks on a \$500 total demo, i.e. ~\$83/desk; this study uses **\$500 per desk** as requested.)

4. Method (and how we limited over-fitting)

- 1. **Walk-forward:** 1m: IS 12 d / OOS 6 d / step 6 d, 3 windows; 5m: IS 60 d / OOS 30 d / step 30 d, 3 windows; 15m/30m: IS 120 d / OOS 45 d / step 45 d, 5 windows; 1h/4h: IS 180 d / OOS 60 d / step 60 d, 8 windows. In each window the config with the best in-sample net P&L (≥ 8 trades) is chosen and then run on the next, unseen window. The stitched out-of-sample (OOS) result is the headline; in-sample numbers are shown only for contrast.
- 2. **Untouched holdout:** the last 42 days (2026-08-23 $\rightarrow$ 2026-10-04; 10 days for 1m) were never used for selection. For each family $\times$ frame we picked the best config on all pre-holdout data and ran it once on the holdout; for the top candidates we also ran the config chosen in the most recent walk-forward window.
- 3. **Multiple testing:** 4,196 grid configs + 10 baselines (grids fixed before looking at results; every one counted). Checks: (a) **random-entry control** — same exits, stops, sizing and costs, random entry times and sides, same trade count, 300 runs per candidate $\rightarrow$ p-value; (b) **deflated Sharpe ratio** (Bailey & López de Prado) with $N = 65$ walk-forward trials; (c) **window concentration** — OOS P&L with the single best window removed; (d) **cost stress** — slippage 10 bps instead of 5.
- 4. **Zero-degree-of-freedom check:** published default parameters (Turtle 20/10, Supertrend 10 $\times$ 3, MACD 12/26/9, etc.) run unchanged — no optimisation, so no selection bias for those specific rules.
- 5. Families explored: the fleet trend, mean-reversion and skew-proxy strategies (with wider stops, R-multiple targets, ADX trend/range regime filters, trailing exits, fewer trades), plus the sourced strategies in §6. Maker-style fills are **not** assumed (the vault is a taker at oracle price).

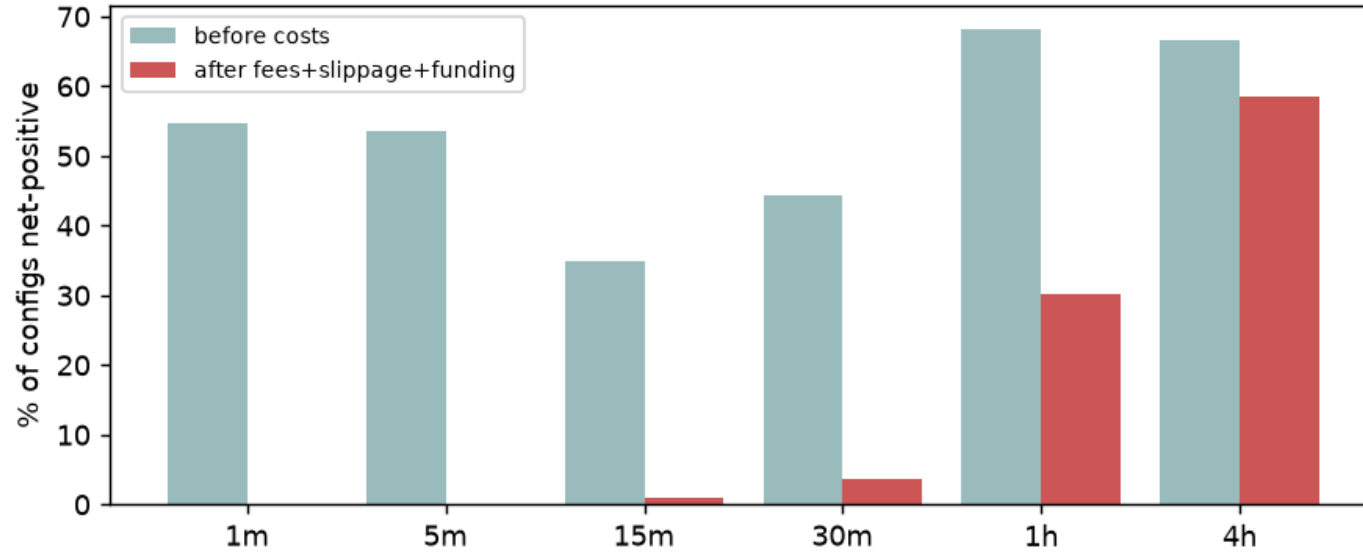
5. Results

5.1 Current demo / fleet desk configs (no optimisation)

desk config	tf	OOS trades	OOS net \$	OOS win %	pre-HO trades	pre-HO net \$	pre-HO PF	pre-HO DD %	10% halts	pre-HO net \$ before costs	HO trades	HO net \$	HO win %	HO DD %
echo (Active 15s mean-rev, evaluated at 1m)	1m	108.0	-92.4	43.0	101	-95.8	0.1	19.2	2	10.5	60	-48.0	37.0	9.8
echo (fleet 15m mean-rev)	15m	36.0	-29.8	44.0	55	-39.8	0.7	11.6	1	4.7	5	-0.4	60.0	1.9
halo (Active 30s mean-rev, evaluated at 1m)	1m	108.0	-100.3	31.0	108	-96.1	0.2	19.2	2	20.2	47	-50.0	36.0	10.3
halo (fleet 1h mean-rev)	1h	22.0	-52.6	18.0	28	-5.4	0.9	10.3	1	0.7	1	0.7	100.0	0.5
lyra (Active 1m skew proxy)	1m	108.0	-93.3	14.0	141	-78.3	0.1	16.0	1	1.0	60	-30.5	27.0	6.7
nova (Active 30m skew proxy)	30m	847.0	-521.9	43.0	1324	-388.2	0.5	77.8	15	53.3	147	-100.7	44.0	20.7
regal (Active 15m trend)	15m	666.0	-497.7	24.0	942	-416.4	0.5	84.1	25	-130.7	114	-100.3	26.0	21.4
regal (fleet 4h trend)	4h	123.0	191.0	30.0	155	556.4	1.9	24.2	6	610.2	9	-36.2	0.0	7.4
vega (Active 5m trend)	5m	313.0	-238.5	22.0	539	-279.0	0.5	56.3	9	-80.5	125	-168.0	13.0	33.9
vega (fleet 1h trend)	1h	417.0	-60.2	30.0	572	191.0	1.1	49.0	17	875.9	35	-36.2	29.0	10.1

Every Active (15s–30m) config loses in OOS (22 of 23 windows) and in the holdout; most lose even before costs or barely break even before costs. The fleet's own 4h trend default was positive pre-holdout (driven by the Nov-2024 and Jul-2025 rallies) but lost in the holdout.

Share of ~700 configs per timeframe that made money (pre-holdout period, simul



5.2 Walk-forward OOS, every family × frame (best IS config re-selected each window)

frame	family	trades	win	avg R	OOS net \$	PF	max DD %	+windows	windows	halts	IS net \$ (selected)
1h	turtle	415.0	27.0	0.2	382.0	1.3	24.2	4.0	8.0	16.0	2233.3
30m	tsmom	137.0	31.0	0.3	240.0	1.6	18.0	1.0	5.0	5.0	788.6
1h	tsmom	300.0	34.0	0.1	145.8	1.2	20.6	3.0	8.0	10.0	2286.9
4h	supertrend	75.0	37.0	0.3	112.9	1.9	5.4	3.0	8.0	0.0	385.7
4h	tsmom	86.0	36.0	0.2	70.3	1.3	8.8	3.0	8.0	0.0	1452.7
1h	supertrend	212.0	37.0	0.0	50.6	1.1	9.8	4.0	8.0	0.0	1027.3
4h	trend	133.0	37.0	0.1	43.5	1.2	7.6	3.0	8.0	0.0	1650.7
15m	skewproxy	350.0	42.0	0.0	27.2	1.1	5.9	2.0	5.0	0.0	12.7
15m	macross	97.0	29.0	0.0	15.1	1.1	14.0	1.0	5.0	1.0	111.2
5m	keltner	218.0	27.0	0.0	8.6	1.0	21.9	2.0	3.0	7.0	216.5
1m	tsmom	58.0	21.0	0.1	8.4	1.1	10.1	1.0	3.0	3.0	-73.6
30m	macross	87.0	25.0	0.0	7.3	1.0	9.9	2.0	5.0	0.0	179.0
15m	tsmom	211.0	35.0	0.0	-1.8	1.0	16.7	2.0	5.0	7.0	317.0
1h	skewproxy	211.0	52.0	-0.0	-2.7	1.0	9.3	5.0	8.0	0.0	337.4
1m	orb	18.0	39.0	-0.1	-8.9	0.8	4.3	1.0	3.0	0.0	-4.7
1h	macross	78.0	31.0	-0.0	-9.3	0.9	5.8	4.0	8.0	0.0	1314.8
30m	supertrend	81.0	31.0	-0.0	-9.9	0.9	6.7	2.0	5.0	0.0	226.0
4h	turtle	80.0	30.0	-0.0	-13.0	0.9	10.2	3.0	8.0	1.0	1370.8
30m	orb	188.0	41.0	-0.0	-14.9	1.0	7.8	2.0	5.0	0.0	-3.9
5m	orb	66.0	33.0	-0.1	-17.8	0.8	7.1	1.0	3.0	0.0	27.7

frame	family	trades	win	avg R	OOS net \$	PF	max DD %	+windows	windows	halts	IS net \$ (selected)
4h	keltner	142.0	31.0	-0.0	-19.2	0.9	10.2	3.0	8.0	1.0	1764.3
4h	skewproxy	91.0	31.0	-0.1	-21.3	0.9	6.9	3.0	8.0	0.0	425.1
1m	skewproxy	108.0	27.0	-0.1	-21.5	0.3	2.0	0.0	3.0	0.0	-32.5
4h	rsi2	106.0	58.0	-0.0	-22.4	0.8	6.8	5.0	8.0	0.0	46.0
15m	bbmr	34.0	50.0	-0.1	-23.5	0.7	4.0	1.0	5.0	0.0	34.3
4h	macd	94.0	30.0	-0.1	-31.7	0.8	6.2	3.0	8.0	0.0	248.6
1m	bbmr	47.0	49.0	-0.2	-39.8	0.5	4.1	0.0	3.0	0.0	-50.2
30m	rsi2	262.0	51.0	-0.0	-40.2	0.6	4.3	1.0	5.0	0.0	-89.5
4h	bbmr	61.0	33.0	-0.1	-42.8	0.8	6.8	2.0	8.0	0.0	298.6
4h	macross	50.0	34.0	-0.2	-43.8	0.6	4.5	2.0	8.0	0.0	300.6
30m	trend	222.0	28.0	-0.1	-57.1	0.8	13.8	2.0	5.0	2.0	231.3
30m	bbmr	74.0	38.0	-0.2	-60.4	0.7	10.1	1.0	5.0	1.0	56.5
5m	turtle	210.0	29.0	-0.1	-69.1	0.9	23.7	1.0	3.0	7.0	-16.2
1m	trend	108.0	23.0	-0.1	-73.4	0.4	7.7	0.0	3.0	0.0	-100.3
1m	supertrend	98.0	32.0	-0.2	-77.0	0.5	9.7	0.0	3.0	0.0	-89.2
5m	trend	273.0	25.0	-0.1	-77.9	0.8	19.4	1.0	3.0	6.0	133.9
1h	rsi2	391.0	54.0	-0.0	-80.4	0.6	4.3	0.0	8.0	0.0	-95.3
1m	turtle	108.0	22.0	-0.2	-87.6	0.4	8.1	0.0	3.0	0.0	-125.1
5m	skewproxy	476.0	34.0	-0.1	-88.6	0.5	7.8	0.0	3.0	0.0	-108.4
5m	supertrend	101.0	29.0	-0.2	-91.2	0.5	13.2	1.0	3.0	1.0	-188.2
15m	orb	197.0	37.0	-0.1	-91.5	0.8	10.5	0.0	5.0	1.0	-10.9
1m	keltner	108.0	21.0	-0.2	-96.4	0.3	8.7	0.0	3.0	0.0	-100.5
5m	macross	108.0	25.0	-0.2	-100.2	0.7	16.0	1.0	3.0	3.0	21.6
1m	rsi2	108.0	9.0	-0.2	-101.1	0.0	8.5	0.0	3.0	0.0	-146.4
1m	macd	108.0	14.0	-0.2	-102.9	0.2	8.1	0.0	3.0	0.0	-145.4
15m	rsi2	490.0	48.0	-0.0	-104.9	0.5	8.7	0.0	5.0	0.0	-279.6
1h	macd	421.0	31.0	-0.1	-106.2	0.8	14.6	4.0	8.0	2.0	159.7
1m	macross	85.0	15.0	-0.3	-112.5	0.2	10.3	0.0	3.0	1.0	-96.3
1h	keltner	389.0	30.0	-0.1	-118.1	0.9	20.6	2.0	8.0	12.0	1952.5
5m	bbmr	111.0	48.0	-0.2	-118.8	0.5	12.0	0.0	3.0	1.0	53.7
30m	skewproxy	342.0	45.0	-0.1	-123.5	0.7	17.1	0.0	5.0	2.0	18.5
30m	macd	398.0	27.0	-0.1	-133.7	0.7	14.8	1.0	5.0	2.0	-160.5
15m	supertrend	137.0	26.0	-0.2	-134.2	0.5	18.5	1.0	5.0	2.0	-116.9
5m	tsmom	194.0	24.0	-0.1	-141.5	0.7	26.2	0.0	3.0	8.0	256.8
1h	orb	403.0	36.0	-0.1	-145.3	0.9	17.2	2.0	8.0	7.0	-38.3
5m	rsi2	374.0	34.0	-0.1	-160.2	0.3	15.4	0.0	3.0	2.0	-266.2
30m	turtle	242.0	31.0	-0.1	-168.0	0.7	19.5	2.0	5.0	7.0	215.0
5m	macd	366.0	25.0	-0.1	-206.4	0.5	19.2	0.0	3.0	4.0	-338.7
1h	trend	313.0	28.0	-0.1	-207.4	0.7	17.8	2.0	8.0	8.0	1366.4
30m	keltner	180.0	22.0	-0.3	-222.7	0.5	24.4	1.0	5.0	4.0	297.7
1h	bbmr	181.0	38.0	-0.3	-229.8	0.6	15.6	0.0	8.0	2.0	214.6
15m	turtle	265.0	26.0	-0.2	-249.8	0.6	23.1	1.0	5.0	11.0	150.6
15m	macd	656.0	23.0	-0.1	-273.4	0.5	19.1	0.0	5.0	5.0	-553.0
15m	trend	335.0	26.0	-0.2	-276.9	0.6	32.1	2.0	5.0	9.0	65.4
15m	keltner	302.0	23.0	-0.2	-335.9	0.6	24.2	0.0	5.0	12.0	-18.8

"Pick the best of everything" walk-forward (each window the optimiser may choose any family on that frame — the most honest summary of the whole search):

frame	trades	OOS net \$	PF	+windows	windows	family picked each window
1m	19.0	-25.1	0.6	0.0	3.0	bbmr; orb; orb
5m	182.0	-77.8	0.8	1.0	3.0	tsmom; tsmom; keltner
15m	169.0	-125.4	0.7	1.0	5.0	bbmr; tsmom; tsmom; tsmom; turtle
30m	169.0	333.4	1.8	2.0	5.0	turtle; turtle; keltner; tsmom; tsmom
1h	373.0	17.7	1.0	3.0	8.0	macross; tsmom; turtle; tsmom; turtle; tsmom; bbmr; keltner
4h	108.0	-81.5	0.7	3.0	8.0	keltner; skewproxy; trend; trend; trend; bbmr; bbmr; trend

5.3 Robustness of the top walk-forward candidates

tf	family	WF OOS \$	+windows	OOS \$ w/o best window	p vs random	random median \$	Sharpe (ann.)	DSR prob	OOS \$ @10bp slip	OOS \$ no costs	HO trades	HO net \$	HO win	HO DD %	config used on holdout
1h	turtle	382.00	4/8	-271.20	0.08	167.50	0.70	0.00	250.10	560.60	39	-77.40	18.00	17.70	{"n_in": 20, "n_out": 10, "stopN": 2, "trail": "none"}
30m	tsmom	240.00	1/5	-188.30	0.01	-15.20	1.00	0.00	226.70	289.00	15	-32.00	20.00	10.60	{"H": 48, "L": 168, "stopAtr": 3, "trailAtr": 0}
1h	tsmom	145.80	3/8	-79.30	0.09	16.70	0.63	0.00	95.50	278.20	21	-50.00	14.00	11.30	{"H": 12, "L": 168, "stopAtr": 3, "trailAtr": 0}
4h	supertrend	112.90	3/8	-13.90	0.08	1.20	1.14	0.00	105.00	128.70	2	-1.30	50.00	1.10	{"adx_min": 25, "m": 2, "n": 20}
4h	tsmom	70.30	3/8	-48.40	0.10	-16.10	0.50	0.00	62.80	72.20	5	-13.00	0.00	3.90	{"H": 12, "L": 500, "stopAtr": 3, "trailAtr": 4}
1h	supertrend	50.60	4/8	-42.90	0.04	-46.60	0.72	0.00	5.60	111.40	21	-6.90	38.00	3.90	{"adx_min": 0, "m": 3, "n": 10}
4h	trend	43.50	3/8	-32.40	0.07	-17.10	0.49	0.00	47.50	70.60	16	-39.40	6.00	8.10	{"adx_min": 0, "breakout": 20, "fast": 9, "slow": 21, "stopAtr": 2, "tpR": 0, "trailAtr": 3}
15m	skewproxy	27.20	2/5	-19.50	0.00	-71.40	1.04	0.00	-18.30	105.80	41	-18.80	27.00	4.10	{"N": 20, "dir": 1, "exitRate": 2e-05, "maxHoldBars": 30, "maxRet": 1.2, "minRate": 0.0001, "minSkew": 10000, "stopPct": 3, "tpPct": 4}

Reading: "OOS \$ w/o best window" is negative for all eight; "p vs random" ≤ 0.05 only for 30m TSMOM (one positive window out of five), 1h Supertrend and the 15m with-flow skew-proxy variant (random 15m entries bleed costs, so beating them is easy; the variant itself is only +\$27 and turns negative at 10 bps slippage) — none of which survives a correction for ~65 trials (Bonferroni threshold ≈ 0.0008); deflated Sharpe probability ≈ 0 for all (the expected best Sharpe from 65 no-skill trials is ≈ 6.6 annualised on this sample, far above any candidate's). **All eight lost money on the holdout.**

5.4 Holdout: best pre-holdout config per family × frame, run once on the last 42 days

frame	family	pre-HO trades	pre-HO net \$ (in-sample!)	pre-HO PF	HO trades	HO net \$	HO win	HO DD %
4h	keltner	142.0	1751.9	3.2	9.0	-28.9	0.0	5.9
1h	keltner	479.0	1560.6	1.6	31.0	-47.7	19.0	13.1
1h	turtle	594.0	1342.4	1.3	39.0	-77.4	18.0	17.7
4h	trend	85.0	1295.7	4.8	2.0	-4.0	0.0	0.9
4h	turtle	132.0	1274.6	2.8	9.0	-22.9	11.0	5.5
1h	tsmom	312.0	1026.1	1.6	21.0	-50.0	14.0	11.3
1h	macross	78.0	1017.1	3.3	5.0	-2.7	40.0	3.1
4h	tsmom	115.0	1000.6	2.5	7.0	-30.1	0.0	7.1
4h	macross	46.0	660.9	4.1	2.0	-7.0	0.0	2.2
4h	supertrend	43.0	621.5	4.9	2.0	-5.2	0.0	1.3
1h	trend	455.0	606.5	1.3	29.0	-29.9	21.0	10.2
30m	tsmom	148.0	531.6	1.9	15.0	-32.0	20.0	10.6
1h	supertrend	303.0	415.3	1.5	21.0	-6.9	38.0	3.9
30m	macross	98.0	118.1	1.5	18.0	-6.7	28.0	5.8
30m	keltner	455.0	90.4	1.1	49.0	-60.7	31.0	14.8
30m	supertrend	117.0	83.1	1.4	13.0	-3.5	31.0	3.8
30m	trend	165.0	70.5	1.2	23.0	-74.4	9.0	15.4

frame	family	pre-HO trades	pre-HO net \$ (in-sample!)	pre-HO PF	HO trades	HO net \$	HO win	HO DD %
4h	bbmr	50.0	56.4	1.5	0.0	0.0	nan	0.0
4h	macd	24.0	40.1	2.5	0.0	0.0	nan	0.0
4h	skewproxy	68.0	36.5	1.2	0.0	0.0	nan	0.0
1h	bbmr	26.0	20.4	1.2	1.0	6.3	100.0	0.5
15m	skewproxy	531.0	19.7	1.0	41.0	-18.8	27.0	4.1
5m	trend	311.0	18.9	1.0	81.0	-119.0	19.0	24.3
15m	macross	85.0	10.2	1.1	15.0	-6.2	40.0	5.3
1h	macd	473.0	4.7	1.0	29.0	-6.3	34.0	3.0
30m	turtle	312.0	3.5	1.0	31.0	-32.2	26.0	10.8
15m	bbmr	48.0	0.6	1.0	4.0	6.7	100.0	0.6
4h	rsi2	88.0	0.2	1.0	6.0	2.1	83.0	0.8
5m	orb	121.0	-1.3	1.0	41.0	-28.0	41.0	8.1
1m	tsmom	98.0	-6.2	1.0	20.0	-17.4	15.0	10.1
30m	bbmr	43.0	-6.5	0.9	2.0	-0.0	50.0	1.0
5m	bbmr	71.0	-13.2	0.9	19.0	-25.0	32.0	6.1
1m	orb	30.0	-19.8	0.7	10.0	-9.1	40.0	3.9
15m	tsmom	280.0	-25.8	1.0	38.0	-101.9	18.0	23.3
1h	skewproxy	340.0	-28.0	0.9	9.0	-9.9	56.0	3.1
5m	keltner	382.0	-31.5	1.0	107.0	-138.6	30.0	32.2
5m	macross	125.0	-35.0	0.7	36.0	-34.9	14.0	7.0
1m	bbmr	52.0	-36.1	0.5	26.0	-25.7	38.0	7.7
1m	macross	65.0	-37.4	0.7	26.0	-43.2	31.0	10.1
1m	skewproxy	180.0	-37.5	0.1	60.0	-17.3	10.0	3.5
30m	skewproxy	355.0	-41.9	0.8	14.0	-6.8	64.0	3.0
1m	supertrend	139.0	-43.3	0.7	37.0	-47.3	16.0	10.1
15m	turtle	301.0	-46.3	0.9	41.0	-52.6	22.0	16.2
1m	trend	95.0	-49.5	0.6	26.0	-29.8	19.0	10.2
5m	tsmom	139.0	-52.7	0.9	42.0	19.5	33.0	13.1
1h	rsi2	293.0	-52.8	0.6	17.0	4.0	76.0	0.8
5m	turtle	197.0	-55.2	0.9	56.0	-129.3	18.0	26.9
1m	turtle	138.0	-56.4	0.6	37.0	-50.1	14.0	10.0
1m	keltner	136.0	-57.7	0.6	25.0	-50.5	4.0	10.1
30m	orb	279.0	-76.2	0.8	38.0	-29.8	39.0	7.6
15m	keltner	374.0	-80.2	0.9	29.0	-58.9	24.0	12.7
15m	trend	370.0	-81.4	0.9	44.0	-24.0	32.0	8.9
30m	rsi2	425.0	-84.2	0.6	44.0	-20.5	50.0	4.9
1m	macd	138.0	-87.7	0.4	51.0	-47.2	20.0	10.1
15m	orb	284.0	-94.3	0.8	37.0	-34.1	30.0	9.2
1m	rsi2	114.0	-95.3	0.0	39.0	-50.6	3.0	10.1
5m	skewproxy	552.0	-98.0	0.6	132.0	-47.7	32.0	10.4
15m	supertrend	153.0	-100.2	0.7	10.0	-16.3	10.0	5.9
30m	macd	564.0	-112.7	0.8	75.0	-40.7	23.0	8.7
5m	supertrend	161.0	-141.5	0.5	47.0	-33.5	30.0	9.9
15m	rsi2	740.0	-148.8	0.5	94.0	-24.9	57.0	5.6
1h	orb	523.0	-157.9	0.8	29.0	-45.2	31.0	12.2

frame	family	pre-HO trades	pre-HO net \$ (in-sample!)	pre-HO PF	HO trades	HO net \$	HO win	HO DD %
5m	rsi2	553.0	-211.6	0.2	154.0	-81.9	30.0	16.6
5m	macd	555.0	-244.1	0.4	152.0	-111.8	18.0	22.4
15m	macd	992.0	-283.2	0.5	123.0	-70.4	18.0	14.7

5 of 65 were positive on the holdout, all by $\leq$ \$20 on $\leq$ 42 trades — consistent with noise. The configs with the largest in-sample profits (+\$1,000-1,750 pre-holdout) all lost on the holdout.

1-USD (Coinbase 1h). Green = walk-forward OOS windows (1h/4h), red = untouched



6. Sourced / published strategies

No third-party code was copied; every strategy was reimplemented from the published rules. Claimed track records (futures in the 1980s–2000s, equities, BTC) were treated as hypotheses and re-tested on XLM with vault costs.

family	source	url	license
trend	Quasaria fleet trend (bot/src/office/strategies.ts): Donchian breakout + EMA filter, ATR stop/trail, EMA exit	in-repo	repo license
bbmr	Quasaria fleet mean-rev (Bollinger 20/2.2 + RSI extreme + ADX<20, target mid band); Bollinger 'Bollinger on Bollinger Bands' (2001)	in-repo / book	repo license; reimplemented
skewproxy	Quasaria Active skew proxy (candle pressure mapped onto funding strategy)	frontend/src/lib/demo/engine.ts	repo license
turtle	Turtle Trading rules (Dennis/Eckhardt; C. Faith 'Way of the Turtle' 2007; public 'Original Turtle Trading Rules' PDF); 20/55-bar Donchian entry, 10/20-bar exit, 2N stop	https://oxfordstrat.com/coasdfsD32/uploads/2016/01/turtle-rules.pdf	rules (pseudocode) only; reimplemented
tsmom	Time-series momentum (Moskowitz, Ooi, Pedersen 2012 JFE 'Time series momentum'; crypto: Liu & Tsyvinski 2021 RFS 'Risks and returns of cryptocurrency')	https://doi.org/10.1016/j.jfineco.2011.11.003	paper; reimplemented
macross	Dual moving-average crossover / Faber 'A Quantitative Approach to Tactical Asset Allocation' (2007) SMA trend filter	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=962461	paper; reimplemented
rsi2	Connors RSI(2) (Connors & Alvarez 'Short Term Trading Strategies That Work' 2008): RSI2 extreme with 200-SMA filter, exit on 5-SMA cross	book	rules only; reimplemented
keltner	Keltner/ATR channel breakout (Chester Keltner 1960; L. Raschke ATR variant)	https://school.stockcharts.com/doku.php?id=technical_indicators:keltner_channels	public description; reimplemented
orb		book	rules only; reimplemented

family	source	url	license
	Opening-range breakout (Toby Crabel 'Day Trading with Short Term Price Patterns and Opening Range Breakout' 1990) adapted to the 00:00 UTC session of 24h crypto		
macd	MACD 12/26/9 signal cross (Gerald Appel) with 200-EMA trend filter	https://school.stockcharts.com/doku.php?id=technical_indicators:moving_average_convergence_divergence_macd	public description; reimplemented
supertrend	Supertrend (Olivier Seban) ATR(10)x3 flip with line as trailing stop	https://www.tradingview.com/support/solutions/43000634738-supertrend/	public description; reimplemented (TradingView built-in, not copied)
chandelier	Chandelier exit (Chuck LeBeau): highest high(22) - 3*ATR(22) trail, used as an exit option on breakouts	https://school.stockcharts.com/doku.php?id=technical_indicators:chandelier_exit	public description; reimplemented
random	Random-entry control: random time + side with each config's own exits/stops/costs	this study	-

Open-source libraries checked (LICENSE files read 2026-10-04): | repo | license | |-----|:-----| | freqtrade/freqtrade-strategies | GPL-3.0 (not used: incompatible to copy; concepts only) | | momentum/backtrader | GPL-3.0 (not used) | | kernc/backtesting.py | AGPL-3.0 (not used) | | jesse-ai/jesse | MIT (permissive; not needed, rules reimplemented) | | QuantConnect/Lean | Apache-2.0 (permissive; not needed, rules reimplemented) |

Published defaults, unchanged (zero degrees of freedom), OOS period = all walk-forward OOS windows (contiguous), then holdout:

strategy	tf	days	trades	win %	avg R	net \$	PF	DD %	Sharpe	cost share	+windows	net w/o best window	p vs random	HO trades	HO win %	HO net \$	HO PF	HO DD %
tsmom (L=168 bars, H=48)	15m	225	174	37.00	0.07	53.80	1.12	16.20	0.71	0.08	2/5	-29.50	0.16	38	18.00	-101.90	0.13	23.30
MA cross 50/200 long-only (Faber-style)	15m	225	50	24.00	-0.11	-28.40	0.75	12.40	-0.64	0.10	2/5	-58.70	0.67	15	40.00	-6.20	0.79	5.30
ORB first 2h UTC	15m	225	179	30.00	-0.18	-142.30	0.70	32.70	-2.37	0.12	1/5	-159.20	0.88	36	28.00	-52.70	0.54	10.50
Keltner 20, 2ATR breakout	15m	225	390	30.00	-0.13	-215.60	0.75	53.70	-2.04	0.19	1/5	-296.90	0.62	76	20.00	-120.90	0.32	30.80
Supertrend 10x3	15m	225	367	26.00	-0.16	-225.80	0.65	47.50	-3.92	0.16	0/5	-204.50	1.00	76	30.00	-61.10	0.56	12.90
trend (fleet default 20/50/20, 2ATR, 3ATR trail)	15m	225	439	27.00	-0.14	-242.10	0.70	54.00	-2.53	0.21	2/5	-276.60	0.71	70	13.00	-132.20	0.22	29.80
turtle S2 (55/20, 2N)	15m	225	276	24.00	-0.23	-247.70	0.63	56.30	-2.74	0.17	1/5	-250.50	0.97	68	29.00	-88.00	0.55	22.60
MACD 12/26/9 + EMA200	15m	225	576	24.00	-0.13	-269.00	0.61	56.70	-4.75	0.25	0/5	-255.40	0.97	119	26.00	-90.50	0.43	18.10
Connors RSI(2) <10 / SMA200	15m	225	786	48.00	-0.10	-273.70	0.43	54.80	-8.10	0.43	0/5	-242.40	0.91	156	45.00	-90.80	0.33	18.30
Bollinger 20/2 + RSI35, no ADX	15m	225	580	44.00	-0.15	-297.00	0.63	63.40	-4.79	0.25	0/5	-284.00	0.88	137	45.00	-82.10	0.69	17.80
turtle S1 (20/10, 2N)	15m	225	453	24.00	-0.19	-297.90	0.68	63.80	-3.40	0.19	0/5	-284.80	0.99	89	20.00	-135.20	0.42	29.50
turtle 20/10 + chandelier	15m	225	558	26.00	-0.17	-313.30	0.60	64.90	-4.50	0.20	0/5	-262.70	1.00	99	16.00	-125.80	0.31	26.30
turtle S1 (20/10, 2N)	1h	480	412	29.00	0.34	541.40	1.28	42.00	1.25	0.07	5/8	-27.10	0.04	39	18.00	-77.40	0.25	17.70
Supertrend 10x3	1h	480	238	36.00	0.25	316.10	1.54	19.70	1.66	0.06	4/8	86.70	0.01	21	38.00	-6.90	0.80	3.90
turtle S2 (55/20, 2N)	1h	480	253	27.00	0.15	114.70	1.17	30.90	0.57	0.07	4/8	-33.20	0.38	21	19.00	-46.70	0.30	10.20
tsmom (L=168 bars, H=48)	1h	480	104	41.00	0.08	36.30	1.15	17.70	0.38	0.04	5/8	11.10	0.07	9	11.00	-20.20	0.25	5.80
MA cross 50/200 long-only (Faber-style)	1h	480	33	24.00	0.18	22.80	1.22	14.90	0.36	0.03	4/8	-47.00	0.26	3	33.00	1.80	1.18	3.20
turtle 20/10 + chandelier	1h	480	539	37.00	0.01	9.20	1.01	28.40	0.18	0.09	3/8	-138.90	0.14	52	31.00	-44.10	0.54	10.50
trend (fleet default 20/50/20, 2ATR, 3ATR trail)	1h	480	398	29.00	-0.01	-73.80	0.91	49.00	-0.15	0.10	2/8	-193.20	0.12	35	29.00	-36.20	0.58	10.10
Connors RSI(2) <10 / SMA200	1h	480	688	60.00	-0.04	-110.50	0.75	24.10	-1.72	0.23	1/8	-111.40	0.06	77	61.00	-21.20	0.67	5.70
Keltner 20, 2ATR breakout	1h	480	349	32.00	-0.06	-118.30	0.85	52.50	-0.49	0.10	2/8	-299.80	0.34	32	38.00	-42.50	0.41	10.10
MACD 12/26/9 + EMA200	1h	480	398	29.00	-0.09	-152.00	0.67	32.50	-2.08	0.13	2/8	-166.00	0.70	36	28.00	-22.70	0.55	5.20
ORB first 2h UTC	1h	480	394	34.00	-0.13	-204.80	0.76	44.60	-1.76	0.11	2/8	-246.60	0.82	35	23.00	-54.90	0.48	11.00
Bollinger 20/2 + RSI35, no ADX	1h	480	585	41.00	-0.12	-261.00	0.76	56.10	-2.08	0.13	1/8	-282.00	0.91	45	42.00	-29.80	0.71	11.20
trend (fleet default 20/50/20, 2ATR, 3ATR trail)	4h	480	120	29.00	0.34	164.60	1.47	24.20	0.79	0.04	3/8	-42.20	0.06	9	0.00	-36.20	0.00	7.40
turtle S2 (55/20, 2N)	4h	480	75	29.00	0.38	106.90	1.45	29.20	0.58	0.03	2/8	-67.70	0.15	4	0.00	-19.80	0.00	4.20
Supertrend 10x3	4h	480	37	38.00	0.46	86.00	2.12	7.50	1.03	0.03	4/8	22.30	0.02	2	0.00	-5.20	0.00	1.30
Keltner 20, 2ATR breakout	4h	480	108	29.00	0.11	35.80	1.13	26.60	0.33	0.04	2/8	-52.60	0.36	8	0.00	-35.60	0.00	7.30
turtle 20/10 + chandelier	4h	480	120	38.00	0.07	32.60	1.15	12.60	0.44	0.05	5/8	-26.10	0.18	9	0.00	-24.00	0.00	4.90
MACD 12/26/9 + EMA200	4h	480	75	37.00	0.03	10.30	1.09	7.10	0.20	0.06	3/8	-12.30	0.43	8	38.00	-7.30	0.34	2.00
MA cross 50/200 long-only (Faber-style)	4h	480	0	nan	nan	0.00	nan	0.00	nan	nan	0/8	0.00	0.47	0	nan	0.00	nan	0.00

strategy	tf	days	trades	win %	avg R	net \$	PF	DD %	Sharpe	cost share	+windows	net w/o best window	p vs random	HO trades	HO win %	HO net \$	HO PF	HO DD %
turtle S1 (20/10, 2N)	4h	480	139	34.00	0.02	-1.90	0.99	28.30	0.16	0.05	5/8	-46.30	0.53	13	15.00	-34.90	0.12	9.00
tsmom (L=168 bars, H=48)	4h	480	1	0.00	-1.00	-5.00	0.00	1.30	-0.64	0.03	0/8	-5.00	0.92	0	nan	0.00	nan	0.00
Connors RSI(2) <10 / SMA200	4h	480	142	63.00	-0.02	-16.50	0.84	7.50	-0.47	0.12	2/8	-31.00	0.36	7	43.00	-1.30	0.49	1.50
Bollinger 20/2 + RSI35, no ADX	4h	480	166	39.00	-0.10	-83.60	0.80	23.30	-0.81	0.06	3/8	-115.30	0.81	12	58.00	12.00	1.46	3.20

On 15m every published rule loses except TSMOM (+\$54, 2/5 windows positive, holdout −\$102). On 1h/4h several are positive OOS, but most lose once the best window is removed, and **none is positive on the holdout except Faber-style 50/200 long-only (3 trades, +\$1.8) and Bollinger 4h (+\$12)**. Connors RSI(2), Bollinger, MACD, Keltner and opening-range breakout are negative on most frames. Supertrend 10×3 on 1h is the strongest (still positive without its best window, p = 0.005), but 36 default checks were run, so p ≈ 0.005 is not significant after correction, and it lost on the holdout.

7. Why the short frames cannot work with these costs

tf	median ATR(14) %	round-trip cost %	cost / ATR	min stop (0.8%) / ATR
30s (est.)	0.09	0.20	2.28	9.12
15s (est.)	0.06	0.20	3.23	12.90
1m	0.12	0.20	1.61	6.45
5m	0.25	0.20	0.79	3.15
15m	0.47	0.20	0.43	1.71
30m	0.68	0.20	0.30	1.18
1h	0.98	0.20	0.20	0.82
4h	1.98	0.20	0.10	0.40

With a 0.8 % minimum stop and 0.2 % round-trip cost, a 1m desk needs a move of ~6 ATR just to reach a 1R target, and costs eat ~1.6 ATR per trade (≈ 20 % of the 1R risk at the 0.8 % minimum stop plus 0.25 % fee buffer). At 1h/4h the cost is 0.1–0.2 ATR — only there can a modest edge survive, and the evidence for one is weak.

8. Oracle-latency check (vulnerability, not a strategy)

The vault fills at the oracle price and accepts prices up to max_price_age (hard cap 3,600 s). Simulation on 45 days of 1m Coinbase data: oracle = market price published every U minutes; a trader opens in the direction of the live market when |market − oracle| exceeds a threshold and closes at the next oracle update.

| oracle update every (min) | entry when |market-oracle| > % | trades | win | avg net (bps, after 20 bps costs) | days | |-----:|-----:|-----:|-----:|-----:|-----:|
| 2.0 | 0.1 | 9312.0 | 0.4 | -2.0 | 45.0 | | 2.0 | 0.2 | 1495.0 | 0.8 | 16.1 | 45.0 | | 2.0 | 0.5 | 129.0 | 1.0 | 58.9 | 45.0 | | 5.0 | 0.1 | 9466.0 | 0.4 | -2.9 | 45.0 | | 5.0 | 0.2 | 3426.0 | 0.7 | 14.7 | 45.0 | | 5.0 | 0.5 | 719.0 | 0.9 | 44.9 | 45.0 | | 10.0 | 0.1 | 5953.0 | 0.4 | -3.4 | 45.0 | | 10.0 | 0.2 | 3413.0 | 0.7 | 13.0 | 45.0 | | 10.0 | 0.5 | 1118.0 | 0.9 | 42.0 | 45.0 | | 15.0 | 0.1 | 4189.0 | 0.4 | -3.4 | 45.0 | | 15.0 | 0.2 | 2888.0 | 0.6 | 11.9 | 45.0 | | 15.0 | 0.5 | 1185.0 | 0.9 | 39.2 | 45.0 |

Above a ~0.25 % gap this is profitable after all costs for any update interval ≥ 2 minutes — i.e. a near-free option against vault LPs. The fleet's oracleGuard only stops the fleet's own entries; it does not protect the vault. Suggested mitigations (for the contracts/keeper owners): push prices on deviation (e.g. > 0.1 %) and at least every 30–60 s; reduce max_price_age well below the 3,600 s cap; reject opens/closes when the oracle is older than N seconds or deviates from a second source; or use delayed / two-step execution or an oracle-age-scaled fee. **None of the strategies in this report uses oracle lag.** Note also that the backtests assume a fresh oracle; a stale oracle would make breakout entries look better than they really are.

9. Recommendation

- 1. No config is robustly net-positive out-of-sample.** Do not describe any desk as profitable; keep "simulated / illustrative" labelling.
- 2. What to deploy to the demo — if anything (proposal only, not applied):** the goal can only be "lose less and look realistic", not "make money".

desk	now (Active)	proposed (research only)	evidence (simulated, \$500 desk)
Vega	5m trend 9/21/10	1h Supertrend 10×3	OOS +\$316/480 d, 238 trades, PF 1.54; holdout −\$6.9 (21 trades)
Regal	15m trend 9/21/10	4h trend, fleet default 20/50/20, 2 ATR stop, 3 ATR trail	OOS +\$165/480 d, 120 trades, PF 1.47; holdout −\$36 (9 trades)
Echo	15s mean-rev		pre-HO ≈ \$0 (48 trades), holdout +\$6.7 (4 trades); OOS negative

desk	now (Active)	proposed (research only)	evidence (simulated, \$500 desk)
		15m mean-rev, strict: BB 20/2.5, RSI 28/72, ADX < 20, 3 ATR stop (trades ~1x/week) or pause	
Halo	30s mean-rev	1h mean-rev, fleet default (BB 20/2.2, RSI 28/72, ADX<20) or pause	OOS −\$53 (22 trades), holdout +\$0.7 (1 trade) — near-idle
Lyra	1m skew proxy	retire skew proxy ; idle, or 1h TSMOM (L 168, H 48, 6 ATR stop) as a labelled experiment	skew proxy (as deployed, contrarian) loses on every frame; only a reversed with-flow 15m variant was positive OOS (+\$27) and it fails the 10 bps slippage stress and the holdout; TSMOM 1h OOS +\$36, holdout −\$20
Nova	30m skew proxy	retire skew proxy ; idle, or 4h Turtle 20/10 + Chandelier exit as a labelled experiment	Nova now: OOS −\$522; Turtle-Chandelier 4h OOS +\$33, holdout −\$24

Trade-off: the 1h/4h desks trade ~0.2–1 times per day, so a fresh demo will look much quieter. 3. **Before any live use:** forward paper-trade the two or three candidates (1h Supertrend, 4h trend, 1h Turtle) for 8–12 weeks with pass/fail rules written down in advance (e.g. net > 0 after costs, PF > 1.2, ≥ 40 trades, beating the random-entry 95th percentile). Fix the oracle-latency issue first.

10. Reproduce

```
cd /workspace/quasaria-strat && python3 -m venv .venv && .venv/bin/pip install numpy pandas numba matplotlib markdown weasyprint tabulate
cd research
../.venv/bin/python fetch_data.py 3600 730    # also: 900 400, 300 200, 60 45  (+ Kraken funding JSON, see report §2)
../.venv/bin/python parity_dump.py 3 && (cd ../bot && npx tsx ../frontend/scripts/parity-backtest.ts vega) && ../.venv/bin/python parity.py
../.venv/bin/python run_wf.py && ../.venv/bin/python analyze.py && ../.venv/bin/python robust.py && ../.venv/bin/python fixed_defaults.py
../.venv/bin/python charts.py && ../.venv/bin/python build_report.py
```

Outputs in results/: windows_<tf>.csv (every config × every window), summary_family_tf.csv, summary_meta.csv, summary_current.csv, wf_folds_selected.csv, robustness_top.csv, fixed_defaults.csv, move_vs_cost.csv, oracle_lag_test.csv, parity.json, tally.json, results.json.

Limitations: one asset and one venue's price history; Coinbase prices stand in for the vault oracle; the 2-year window contains a few very large trend moves that dominate trend results; 1m history is only 45 days; sub-minute frames are approximated; fleet-level caps and drift halts not modelled; the 10 % halt is modelled as a 7-day operator reset. All results are simulated.